

Lectures on Multiphysics Modeling

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Fluids 2

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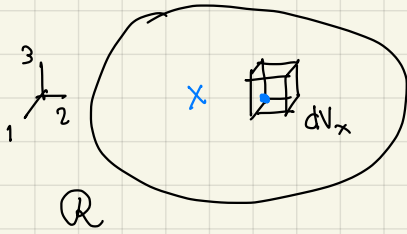
Three conceptual steps common to all continuum mechanics models/theories

- Kinematics (geometry of the unknowns) ✓
- Balance laws (e.g. $m\mathbf{a} = \mathbf{F}$, balance of linear momentum)
- Constitutive equations (how stresses depend on the motion)

To write balance laws we need to introduce forces and stress.

Forces

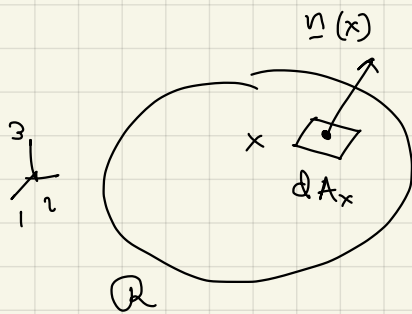
(a) volume or body forces



$\underline{b}$, force per unit volume [quantity: $\underline{b} = \underline{sg}$]

Contribution to total force: $\int_R \underline{b}(x) dV_x$

(b) surface or contact forces



$\underline{n}(x)$ outer unit normal at $x \in \partial R$

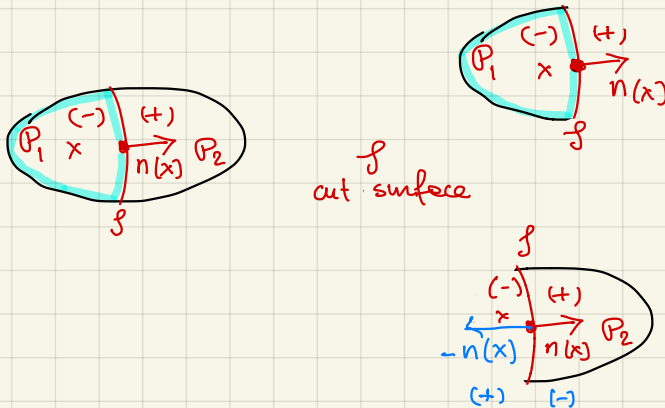
$\underline{t}_{n(x)}(x)$ force per unit area [pressure: $\underline{t}_{n(x)}(x) = -p(x) \underline{n}(x)$]

Contribution to total force: $\int_{\partial R} \underline{t}_{n(x)} dA_x$

(b)' contact forces like (b) act also in the interior and they may be revealed by cut surfaces (Cauchy hypothesis)

$\underline{n}(x)$ outward unit normal, it identifies a (+) side and a (-) side

$\underline{t}_{n(x)}$ is the force p.u. area that the material on the (+) side exerts on the material on the (-) side



cut surface

$$\int_S \underline{t}_{n(x)} dA_x$$

is the contact force that P_2 exerts on P_1

$\int_S \underline{t}_{-n(x)} dA_x$ is the contact force that P_1 exerts on P_2

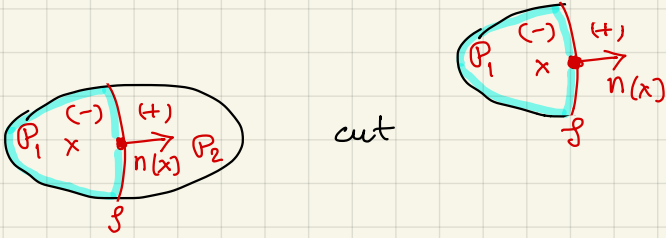
Stress tensor

The stress tensor (3×3 symmetric matrix) allows to compute the contact force $\underline{t}_n$ per unit area at any point of the continuum exchanged through the cut surface of normal $\underline{n}$ as

$$\underline{t}_n = \underline{T} \underline{n} \quad (\text{Cauchy theorem})$$

$\underline{n}(x)$ outward unit normal, it identifies a (+) side and a (-) side

$\underline{T}(x) \underline{n}(x)$ is the force p.u. area that the material on the (+) side exerts on the material on the (-) side



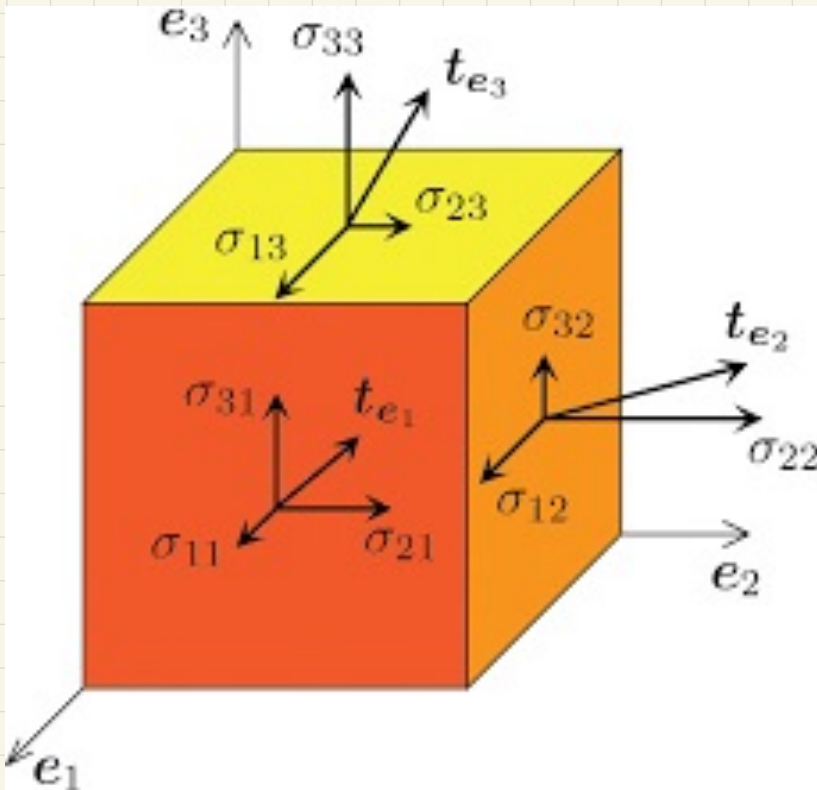
$$\int_S \underline{T}(x) \underline{n}(x) dA_x$$

is the contact force that P_2 exerts on P_1

$$\underline{t}_n = \underline{T} \underline{n} \quad \text{or} \quad \begin{bmatrix} t_{n1} \\ t_{n2} \\ t_{n3} \end{bmatrix} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

$$T_{ij} = \underline{T} \underline{e}_j \cdot \underline{e}_i$$

$$\underline{t}_{e_1} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} T_{11} \\ T_{21} \\ T_{31} \end{bmatrix}; \quad T_{21} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} T_{11} \\ T_{21} \\ T_{31} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \underline{T} \underline{e}_1 \cdot \underline{e}_2$$



$\underline{T}$ is symmetric $T_{ji} = T_{ij}$

(6 independent components)

$\Rightarrow$ A 6-vector in Voigt notation

$$\left\{ \begin{bmatrix} T_{11} \\ T_{22} \\ T_{33} \\ T_{23} \\ T_{13} \\ T_{12} \end{bmatrix} \right\} \quad \text{or} \quad \left\{ \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{yz} \\ \tau_{xz} \\ \tau_{xy} \end{bmatrix} \right\}$$

normal
components/stresses

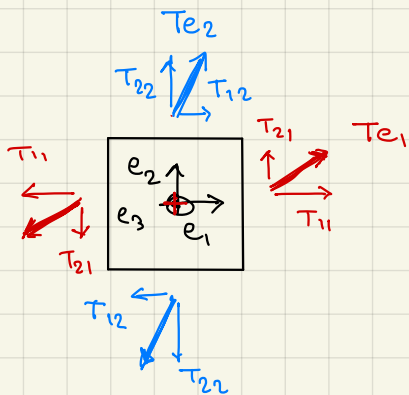
tangential (shear)
components/stresses

Symmetry of T

Assume T constant for simplicity. $T(-\underline{e}_1) = -T\underline{e}_1$, same for $\underline{e}_2, \underline{e}_3$

Elementary cube of edge length $h \ll 1$, centered at the origin

$$t_{-n} = T(-n) = -Tn = -t_n$$



$$T_{21}h - T_{12}h = 0 \text{ for rot. equil.}$$

To leading order in h , the component along x_3 of the moment with respect to the origin is $(T_{21} - T_{12})h$. Equilibrium (or balance of angular momentum) requires this to vanish, so $T_{21} = T_{12}$.

Similar argument for other indices leads to

$$T_{ji} = T_{ij} \quad (\text{symmetry of the stress tensor})$$

=

Notation

$$\underline{T} \underline{a} \cdot \underline{b} = [T][a] \cdot [b] = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\begin{bmatrix} T_{11}a_1 + T_{12}a_2 + T_{13}a_3 \\ T_{21}a_1 + T_{22}a_2 + T_{23}a_3 \\ T_{31}a_1 + T_{32}a_2 + T_{33}a_3 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \left(\sum_{i,j=1}^3 \right) T_{ij} a_j b_i$$

Alternative in matrix product notation $\underline{T} \underline{a} \cdot \underline{b} = [b]^T [T][a] =$

$$= \begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix} \begin{bmatrix} T_{11}a_1 + T_{12}a_2 + T_{13}a_3 \\ T_{21}a_1 + T_{22}a_2 + T_{23}a_3 \\ T_{31}a_1 + T_{32}a_2 + T_{33}a_3 \end{bmatrix}$$

$1 \times 3 \qquad 3 \times 1$

Total force on an arbitrary part P

$$\int_{\partial P} \underline{T} \underline{n} dA + \int_P \underline{b} dV$$

[Transform integral balance laws into pointwise differential equations (localization)]

[Main tool is divergence theorems that transform integrals over the boundary into volume integrals involving derivatives]

Divergence theorems

$$\int_{\partial R} \varphi \underline{n} dA = \int_R \nabla \varphi dV$$

$$\int_{\partial R} \underline{v} \cdot \underline{n} dA = \int_R \operatorname{div} \underline{v} dV$$

where the scalar $\operatorname{div} \underline{v}$ is given by $\operatorname{div} \underline{v} = \sum_i \frac{\partial}{\partial x_i} v_i$

$$\int_{\partial R} \underline{S} \underline{n} dA = \int_R \operatorname{div} \underline{S} dV$$

where the vector $\operatorname{div} \underline{S}$ is given by $(\operatorname{div} \underline{S})_i := S_{ij,j} = \sum_j \frac{\partial}{\partial x_j} S_{ij}$

Proof of last identity. Take $\underline{a}$ arbitrary constant vector and use $\int_{\partial R} \varphi n_j dA = \int_R \frac{\partial}{\partial x_j} \varphi dV$

$$\underline{a} \cdot \int_{\partial R} \underline{S} \underline{n} dA =$$

$$\int_{\partial R} \varphi n_j dA = \int_R \frac{\partial}{\partial x_j} \varphi dV$$

$$= \underline{a} \cdot \int_R \operatorname{div} \underline{S} dV \quad \forall \underline{a}$$

$$= \int_{\partial R} a_i S_{ij} n_j dA = \int_R \frac{\partial}{\partial x_j} (a_i S_{ij}) dV = \int_R a_i \frac{\partial}{\partial x_j} S_{ij} dV =$$

Equality for every $\underline{a} \Rightarrow$ one equal. ✓

We have concluded

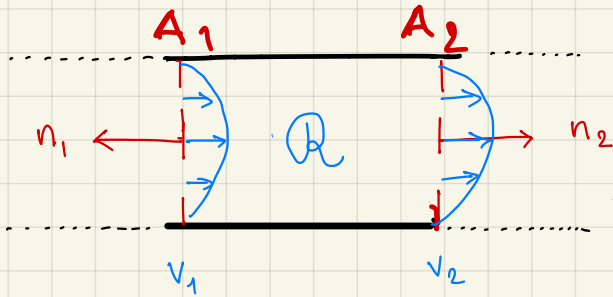
$$\int_{\partial P} \underline{T} \underline{n} dA + \int_P \underline{b} dV = \int_P (\operatorname{div} \underline{T} + \underline{b}) dV$$

$$\int_{\partial P} \underline{v} \cdot \underline{n} dA = \int_P \operatorname{div} \underline{v} dV$$

For an incompressible fluid the net volume flux out of a closed region is always zero

$$\text{Incompressibility} \Rightarrow \int_{\partial R} \underline{v} \cdot \underline{n} dA = 0 = \int_R \text{div } \underline{v} dV \quad \forall R \Rightarrow \text{div } \underline{v} = 0$$

Simplest example: pipe geometry



$\underline{v} \cdot \underline{n} = 0$ at lateral boundary (impermeable wall)

$$\underbrace{\int_{A_1} \underline{v}_1 \cdot \underline{n}_1 dA}_{< 0} + \underbrace{\int_{A_2} \underline{v}_2 \cdot \underline{n}_2 dA}_{> 0} = 0$$

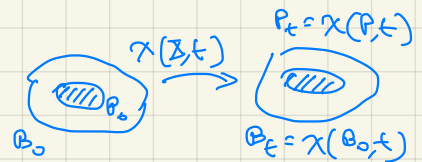
$$\int_{A_1} \underline{v}_1 \cdot \underline{n}_1 dt dA + \int_{A_2} \underline{v}_2 \cdot \underline{n}_2 dt dA$$

With dt : volume flowing through cross section A_1 in time dt

without dt : " " " " " " A_1 per unit time = volume discharge (portata)

For a flow of an incompressible fluid through a (rigid) pipe it's constant.

Remark $\text{div } \underline{v} = 0$ really comes from the statement $\frac{d}{dt} \text{vol}(P_t) = 0$



$$\frac{d}{dt} \int_P 1 dV_x = \frac{d}{dt} \int_{P_0} 1 J dV_{\bar{x}} = \int_{P_0} \dot{J} dV_{\bar{x}} = \int_{P_0} J \text{div } \underline{v} \Big|_{\bar{x} = \xi(x,t)} dV_{\bar{x}} = \int_P \text{div } \underline{v} dV_x = 0 \quad \forall P$$

$$\Rightarrow \text{div } \underline{v} = 0$$

Constitutive equation for Newtonian fluids

$\underline{T}$, Cauchy stress, symmetric from balance of angular momentum

Incompressible Newtonian fluid (def.) is one governed by

$$\underline{T} = -p \underline{I} + \underline{\sigma} (\nabla_x \underline{v}) \quad \text{with } \underline{\sigma}(\cdot) \text{ a linear function of } \nabla_x \underline{v}.$$

$$\underline{T} = -p \underline{I} + 2\mu \underline{D} \quad \text{or} \quad \underline{T} \underline{n} = -p \underline{n} + 2\mu \underline{D} \underline{n}$$

$\mu > 0$ viscosity

$$\underline{I} \text{ identity } (\underline{I})_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{else} \end{cases}$$

$$\underline{D} = \text{sym}(\nabla_x \underline{v}) =$$

$$= \frac{1}{2} (\nabla_x \underline{v} + \nabla_x^T \underline{v}) \quad \text{or} \quad D_{ij} = \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) = 0 \text{ if } \underline{v} \equiv 0$$

↓

$$[\underline{T} = -p \underline{I} \text{ or } \underline{T} \underline{n} = -p \underline{n} \text{ in hydrostatics}] \quad \leftarrow \quad [\text{hydrostatics}]$$

Remark $\text{sym } A = \frac{1}{2} (A + A^T), \quad \text{skw } A = \frac{1}{2} (A - A^T)$

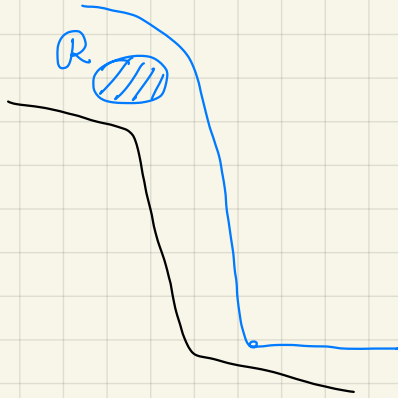
In the eq. above, the skew symmetric part of $\nabla \underline{v}$ does not appear. As we will see later, this is because $\text{skw}(\nabla \underline{v})$ is related to rigid rotations

$$[(\text{div } \underline{T})_i = T_{ij,j} \quad \text{with Einstein's summation convention}]$$

$$\begin{aligned} (\text{div } \underline{T})_i &= \sum_j \frac{\partial}{\partial x_j} T_{ij} \\ &= \sum_j \frac{\partial}{\partial x_j} \left(-p \delta_{ij} + \mu \frac{\partial v_i}{\partial x_j} + \mu \frac{\partial v_j}{\partial x_i} \right) \\ &= -\frac{\partial}{\partial x_i} p + \mu \sum_j \frac{\partial^2 v_i}{\partial x_j^2} + \underbrace{\mu \frac{\partial}{\partial x_i} \sum_j \frac{\partial v_j}{\partial x_j}}_{= \text{div } \underline{v} = 0} \\ &= \left(-\nabla p + \mu \Delta \underline{v} \right)_i \end{aligned}$$

$$\sim (\Delta \underline{v})_i := \Delta(v_i) = \sum_j \frac{\partial^2}{\partial x_j^2} v_i \quad \text{Laplacian of a vector field}$$

Now we can conclude by writing $F = ma$ for an arbitrary part of the region of space occupied by fluid flow.



$$\int_R \rho a \, dV = \int_{\partial R} T n \, dA + \int_R b \, dV$$

$$= \int_R (\text{div } T + b) \, dV$$

Equality $\forall R \Rightarrow$ the integrands are the same

$$a = \dot{v} = \frac{\partial v}{\partial t} + (\nabla v) v = \frac{\partial v}{\partial t} + (v \cdot \nabla) v$$

$$div T = -\nabla p + \mu \Delta v$$

Get Navier-Stokes system for an incompressible fluid

$$\begin{cases} \rho \left(\frac{\partial \underline{v}}{\partial t} + (\nabla \underline{v}) \underline{v} \right) = -\nabla p + \mu \Delta \underline{v} \\ div \underline{v} = 0 \end{cases}$$

Remark The equations determine p only up to an additive constant since they only depend on ∇p . Fixing the pressure at one point of the boundary (e.g., atmospheric pressure at a place exposed to the atmosphere) removes the degeneracy.

Typical boundary conditions for N-S equations

- Inlet either prescribe all velocity components
or pressure p_i and zero tangential velocity

$$\underline{T} \underline{n} \cdot \underline{n} = -p_i, \quad \underline{v} \cdot \underline{t} = 0 \quad (\underline{t} \text{ tangent}, \underline{t} \cdot \underline{n} = 0)$$

- Outlet vanishing tangential stress, with prescribed pressure p_o

$$\underline{T} \underline{n} = -p_o \underline{n}$$

- Symmetry no penetration and vanishing shear stress

$$\underline{v} \cdot \underline{n} = 0, \quad \underline{T} \underline{n} \cdot \underline{t} = 0$$

- No-slip prescribed velocity, equal to the wall velocity

$$\underline{v} = \underline{v}_w \quad (\underline{v}_w = 0 \text{ for a fixed wall})$$